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SEMICONDUCTOR DEVICES WITH LOW LEAKAGE CURRENT AND METHODS OF FABRICATING THE SAME

Abstract

Semiconductor devices and methods are provided. In an embodiment, a method includes providing a workpiece including a first hard mask layer on a top surface of a substrate, performing an ion implantation process to form a doped region in the substrate, after the performing of the ion implantation process, annealing the workpiece at temperature T1. The method also includes selectively removing the first hard mask layer, after the selectively removing of the first hard mask layer, performing a pre-bake process at temperature T2, and, after the performing of the pre-bake process, epitaxially growing a vertical stack of alternating channel layers and sacrificial layers on the substrate, where the temperature T2 is lower than the temperature T1.

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Background/Summary

PRIORITY DATA [0001] The present application is a continuation application of U.S. patent application Ser. No. 17/834,202, filed Jun. 7, 2022, which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0003] For example, as integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate metal-oxide-semiconductor field effect transistor (multi-gate MOSFET, or multi-gate devices) have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or a portion thereof, disposed over more than one side of a channel region. Multi-bridge-channel (MBC) transistors are examples of multi-gate devices that have become popular and promising candidates for high performance and low leakage applications. An MBC transistor has a gate structure that can extend, partially or fully, around a channel region to provide access to the channel region on two or more sides. Because its gate structure surrounds the channel regions, an MBC transistor may also be referred to as a surrounding gate transistor (SGT) or a gate-all-around (GAA) transistor. While existing MBC transistor structures are generally adequate for their general purposes, they are not satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates a flowchart of an exemplary method for fabricating a semiconductor device, according to various embodiments of the present disclosure.

[0006] FIGS. 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, 15A, 16A and 17A (FIGS. 2A-17A) illustrate fragmentary cross-sectional views of an exemplary workpiece during

various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

[0007] FIG. 7C shows a block diagram of a tool used during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

[0008] FIGS. 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B, 15B, 16B and 17B (FIGS. 2B-17B) illustrate fragmentary cross-sectional views of an exemplary workpiece during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

[0009] FIG. 18 illustrates a flowchart of an exemplary method for determining and fine-tuning parameters of thermal treatments involved in the fabrication stages in the method of FIG. 1, according to various embodiments of the present disclosure.

[0010] FIG. 19A illustrates a bar chart showing the relationships among parameters of the thermal treatments, according to various embodiments of the present disclosure.

[0011] FIG. 19B illustrates a table showing ranges of parameters of the thermal treatments, according to various embodiments of the present disclosure.

[0012] FIG. 20 illustrates doping profiles of a doped region of the workpiece before and after performing an anneal process.

[0013] FIG. 21 illustrates a doping profile of the doped region after performing a pre-bake process.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0017] Formation of an MBC transistor includes formation of a stack that includes a number of

channel layers interleaved by a number of sacrificial layers over a substrate, where the sacrificial layers may be selectively removed to release the channel layers as channel members. The stack and a portion of the substrate are patterned to form active regions. A gate structure that includes a dielectric layer and a conductive layer is then formed to wrap around and over each of the channel members. However, in some instances, MBC transistors may suffer current leakage near the patterned portion (i.e., mesa region or mesa structure) of the substrate. More specifically, n-type MBC transistors may be formed in and over a p-type well (e.g., boron-doped p well) in the substrate, p-type MBC transistors may be formed in and over an n-type well (e.g., phosphorous-doped n well) in the substrate. Due to the implementation of some thermal treatments (e.g., annealing, oxidation) during the formation of the n-type MBC transistors and p-type MBC transistors, dopants (e.g., phosphorous) in the n-type well of the p-type MBC transistors may diffuse into the p-type well of the n-type MBC transistors, reducing the dopant concentration in the mesa region of the n-type MBC transistors, thereby increasing the junction leakage. As the separation spacing of two adjacent active regions becomes smaller, the undesired diffusion may be more severe. Although increasing an implant dosage of an ion implantation process for forming the n-type well may increase the dopant concentration in the mesa region, the number of dopants that are available to diffuse during the thermal treatments also increase, leading to a further degraded device performance.

[0018] The present disclosure provides methods for forming semiconductor devices with reduced leakage current. In an embodiment, a method includes providing a workpiece including a first hard mask layer on a top surface of a substrate, performing an ion implantation process to form a doped region in the substrate, after the performing of the ion implantation process, annealing the workpiece at temperature T1. The method also includes selectively removing the first hard mask layer, after the selectively removing of the first hard mask layer, performing a pre-bake process at temperature T2, and, after the performing of the pre-bake process, epitaxially growing a vertical stack of alternating channel layers and sacrificial layers on the substrate, where the temperature T2 is lower than the temperature T1.

[0019] The various aspects of the present disclosure will now be described in more detail with reference to the figures. In that regard, FIG. 1 is a flowchart illustrating method 100 of forming a semiconductor device according to embodiments of the present disclosure. Method 100 is described below in conjunction with FIGS. 2A-17A and FIGS. 2B-17B, which are fragmentary cross-sectional views of a workpiece 200 at different fabrication stages, FIG. 7C, which is a block diagram of tools used at different fabrication stages, and FIGS. 19A-19B, 20 and 21, which are parameters of thermal treatments at different fabrication stages and the mesa region's doping profiles, according to embodiments of method 100. FIGS. 2B-17B depict cross-sectional views of the workpiece 200 taken along line B-B' shown in FIGS. 2A-17A during various fabrication stages in the method 100. Method 100 is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated therein. Additional steps may be provided before, during and after the method 100 and some steps described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all steps are described herein in detail for reasons of simplicity. Because the workpiece 200 will be fabricated into a semiconductor device 200 upon conclusion of the fabrication processes, the workpiece 200 may be referred to as the semiconductor device 200 as the context requires. For avoidance of doubts, the X, Y and Z directions in FIGS. 2A-17A and FIGS. 2B-17B are perpendicular to one another and are used consistently throughout FIGS. 2A-17A and FIGS. 2B-17B. FIG. 18 is a flowchart illustrating a method 300 of determining and fine-tuning parameters of thermal treatments involved in several fabrication stages of the method of FIG. 1, according to embodiments of the present disclosure. Method 300 is described below in conjunction with FIGS. 2A-17A and FIGS. 2B-17B and further in conjunction with FIGS. 19A-19B, 20 and 21, according to embodiments of method 300. Method 300 is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated therein. Additional

steps may be provided before, during and after the method **300** and some steps described can be replaced, eliminated, or moved around for additional embodiments of the method. Throughout the present disclosure, like reference numerals denote like features unless otherwise excepted.

[0020] Referring to FIGS. **1**, **2A** and **2B**, method **100** includes a block **102** where a first hard mask layer **204** is formed on a top surface **202t** of a substrate **202**. In an embodiment, the substrate **202** is a bulk silicon substrate (i.e., including bulk single-crystalline silicon). The substrate **202** may include other semiconductor materials in various embodiments, such as germanium, silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or combinations thereof. In some alternative embodiments, the substrate **202** may be a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GeOI) substrate, and includes a carrier, an insulator on the carrier, and a semiconductor layer on the insulator.

[0021] The first hard mask layer **204** is then formed on the top surface **202t** of the substrate **202**. In embodiments represented in FIGS. **2A-2B**, the first hard mask layer **204** is a single-layer structure and includes a pad oxide layer formed by using a first process **206**. The first process **206** may be a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a thermal oxidation process, or other suitable processes. In an embodiment, the first process **206** includes a thermal oxidation process and the pad oxide layer includes silicon oxide. The thermal oxidation process may be performed under 1 atmosphere (atm) and at a process temperature TO ranging from about 600° C. to about 1200° C. in a process environment that may include one or more reaction gases such as H.sub.2, O.sub.2 and/or 1,2-Dichloroethylene (DCE) for a duration t0 of several hours with a flow rate of about 0.1 standard liter per minute (slm) to about 10 slm. Carrier gases such as N.sub.2, Ar, He may be used to transport the reaction gases into a furnace or a process chamber. The process temperature TO is designed such that metal contamination on the top surface **202t** of the substrate **202** may be removed and metal gatherings may be reduced during subsequent fabrication stages. The first hard mask layer **204** may provide protection for the substrate **202** during subsequent fabrication stages. In some other implementations, the first hard mask layer **204** may be a multi-layer structure and may further include a pad nitride (e.g., silicon nitride) layer formed on the pad oxide layer.

[0022] After forming the first hard mask layer **204**, a patterned photoresist layer may be formed on the first hard mask layer **204**, a first etching process may be applied to pattern the first hard mask layer **204**, and a second etching process may be then conducted to remove a portion of the substrate **202** exposed by the first hard mask layer **204** to form a trench. The trench may be used as an alignment mark during subsequent fabrication stages. After the forming of the trench, the pad nitride layer may be selectively removed, leaving the pad oxide layer on the top surface **202t** of the substrate **202** according to some embodiments.

[0023] Referring to FIGS. **1**, **3A**, **3B**, **4A** and **4B**, method **100** includes a block **104** where one or more ion implantation processes are performed to form an N-type well and/or a P-type well in the substrate **202**. After forming the first hard mask layer **204**, a mask layer **208** is formed over the substrate **202**, such as on the first hard mask layer **204**. The mask layer **208** may include one or more material layers, such as a bottom layer **208b**, a middle layer **208m**, and a top photoresist layer **208t**. In some embodiments, the bottom layer **208b** may include a hard mask material such as a carbon-containing material. The middle layer **208m** may include anti-reflective materials (e.g., a backside anti-reflective coating (BARC) layer) to aid the exposure and focus of the top photoresist layer **208t**.

[0024] The mask layer **208** may be then patterned to expose a first portion **202a** of the substrate **202** while a second portion **202b** of the substrate **202** is covered, as exemplary shown in FIGS. **4A** and **4B**. The first hard mask layer **204** covers both the first portion **202a** and the second portion **202b** of the substrate **202** as an implantation screen layer. P-type MBC transistors may be formed

on the first portion **202a** and n-type MBC transistors may be formed on the second portion **202b**. With the patterned mask layer **208** as an implant mask, an ion implantation process **210** is performed to form an N-type doped region **202N** in the first portion **202a** of the substrate **202**. If the implant dosage of the implantation process **210** is too high, although a mesa region (e.g., mesa region **500T** shown in FIG. **17A**) may maintain a satisfactory dopant concentration after a number of thermal treatments, more dopants may diffuse into adjacent devices and/or adjacent features, leading to worse device performance. If the implant dosage of the implantation process **210** is too low, the mesa region may not have a sufficient and satisfactory dopant concentration, leading to an increased leakage current.

[0025] The n-type doped region **202N** may include n-type dopants, such as phosphorus (P), arsenic (As), other n-type dopants, or combinations thereof. A depth of the n-type doped region **202N** may be between about 300 nm and about 500 nm, and a dopant concentration of the n-type doped region **202N** may be not uniform throughout its depth along the Z axis. A doping profile of the n-type doped region **202N** after the implantation process **210** is shown in FIG. **20** and is represented by curve **410**. For example, in the present embodiment, a top portion **202NT** of the n-type doped region **202N** may have a dopant concentration **C1** greater than about $1\text{E}19$ atoms/cm³. A depth **D1** (from the top surface **202t** of the substrate **202**) of the top portion **202NT** may be between about 30 nm and about 50 nm. A dopant concentration **C2** of a bottom portion **202NB** of the n-type doped region **202N** is less than the dopant concentration **C1** and is less than $1\text{E}19$ atoms/cm³. In some embodiments, the n-type doped region **202N** may include or may be a part of an n-well structure. After the implantation process **210**, the patterned mask layer **208** may be selectively removed. Similar operations (e.g., forming a patterned mask layer to expose the second portion **202b** of the substrate **202** while the first portion **202a** of the substrate **202** is covered and then performing an ion implantation process) may be performed to form a p-type doped region **202P** (shown in FIG. **5A**) in the second portion **202b** of the substrate **202**. The p-type doped region **202P** may include p-type dopants, such as boron (B), boron difluoride (BF₂), other p-type dopant, or combinations thereof. The p-type doped region **202P** may include or may be a part of a p-well structure. In some embodiments, a doping profile of the p-type doped region **202P** may be in a way similar to the doping profile of the n-type doped region **202N**. In some other implantations, instead of performing the ion implantation process **210**, a diffusion process and/or other suitable doping process may be used to form the n-type doped region **202N** and the p-type doped region **202P**. It is understood that the p-type doped region **202P** may be formed before forming the n-type doped region **202N**, in some other embodiments.

[0026] Referring to FIGS. **1**, **5A** and **5B**, method **100** includes a block **106** where an anneal process **212** is performed to the workpiece **200** after the performing of the ion implantation process **210**. The anneal process **212** is performed using a thermal budget, which defines a total amount of thermal energy transferred to the workpiece during the given elevated temperature operation, such as a product of a process temperature and a process duration (i.e., process temperature multiplied by the process duration). The total thermal budget associated with a final semiconductor device **200** is an accumulation of thermal energies transferred to the intermediate workpiece **200** through all thermal processes, such as the summation of thermal budgets through all thermal processes, which cannot be greater than the allowable total thermal budget. The anneal process **212** may include a rapid thermal process (RTP), a laser spike anneal process, a flash anneal process, a furnace anneal process, other suitable thermal process, or combinations thereof. The anneal process **212** may include an anneal temperature **T1** ranging from about 950° C. and to about 1200° C. such that the anneal process **212** may activate dopants in the n-type doped region **202N** and/or dopants in the p-type doped region **202P** and also facilitate movement of the dopants from interstitial sites to substitutional lattice sites to reduce dislocations in crystal lattice and repair defects caused by the ion implantation process **210**. Due to the thermal budget and the anneal temperature of the anneal process **212**, the anneal temperature **T1** may be maintained for a duration **t1** measured by seconds

or microseconds. For example, the duration t_1 of the anneal process **212** may be less than about 60 seconds. The anneal process **212** may be performed under 1 atmosphere (atm) and in a gas environment that includes gases such as oxygen, helium (He), nitrogen (N.sub.2), argon (Ar), hydrogen (H.sub.2), combinations thereof, and/or other suitable gases. In embodiments where the gas environment includes multiple gases, a flow rate of nitrogen (N.sub.2) or Argon (Ar) may be between about 10 slm and 100 slm. After the anneal process **212**, a doping profile of the n-type doped region **202N** may change due to thermal diffusion caused by the anneal process **212**. The doping profile of the n-type doped region **202N** is shown in FIG. **20** and is represented by curve **420**. More specifically, the dopant concentration of at least a part of the top portion **202NT** of the n-type doped region **202N** may decrease, the dopant concentration of the bottom portion **202NB** of the n-type doped region **202N** may increase, compared to the doping profile of the n-type doped region **202N** (represented by curve **410** shown in FIG. **20**) before the anneal process **212**.

[0027] Referring to FIGS. **1**, **6A** and **6B**, method **100** includes a block **108** where the first hard mask layer **204** is selectively removed by an etching process **214** (e.g., a wet etching process) without substantially damaging the substrate **202**. In embodiments where the first hard mask layer **204** includes the pad oxide layer, a dilute HF dip process may be performed to remove the first hard mask layer **204**. It is understood that the etching process **214** may be performed in an etching tool **215** (shown in FIG. **7C**) that is different from the tool used for the anneal process **212**.

[0028] Referring to FIGS. **1**, **7A**, **7B** and **7C**, method **100** includes a block **110** where a pre-bake process **218** is performed to the workpiece **200**. After the performing of the etching process **214**, the workpiece **200** may be transferred from the etching tool **215** to an integrated process system **216**. The integrated process system **216** may be a cluster tool (such as a CVD tool) including multiple process chambers (e.g., process chambers **216a**, **216b**, **216c**) configured to conduct various fabrication processes therein and one or more robot arms **216r** for motion. It is understood that, after the etching process **214**, there may be residues of the first hard mask layer **204** left on the top surface of the substrate **202**, and during the transportation of the workpiece **200** from the etching tool **215** to the integrated process system **216**, the exposed top surface **202t** of the substrate **202** may be oxidized. In addition, the performing of the etching process **214** may cause a moist substrate **202**. Therefore, after being transferred to the integrated process system **216** (e.g., CVD tool), the workpiece **200** may be placed in a first process chamber **216a** of the integrated process system **216** and an in-situ preclean process may be performed. In some embodiments, the in-situ preclean process may be a chemical oxide removal process that implements a combination of fluorine-containing gas (e.g., NF₃, HF, and/or other suitable fluorine-containing gases) and catalytic gas (e.g., NH₃, H₂O, alcohol, and/or other suitable gases). After the performing of the preclean process in the first process chamber **216a**, the workpiece **200** may be transferred by the robot arm **216r** from the first process chamber **216a** to a second process chamber **216b** (shown in FIG. **7C**), and a pre-bake process **218** is then performed in the second process chamber **216b** at temperature T_2 to evaporate the solvent in the substrate **202** for a time duration t_2 that is sufficient to dry the workpiece **200** as well as remove any undesirable components (i.e., oxide residue and/or unintentionally formed oxidized layer) on the top surface **202t** of the substrate **202**.

[0029] In an embodiment, the pre-bake process **218** is performed after the removal of the first hard mask layer **204** and before the epitaxial growth of semiconductor layers. The thermal budget of the pre-bake process **218** may be less than the thermal budget of the anneal process **212**. In some other technologies, the pre-bake process **218** may be performed at temperature T_2 that may be higher than the anneal temperature T_1 to further reduce the defects caused by the ion implantation process **210**. However, in the present disclosure, the temperature T_2 is lower than the anneal temperature T_1 to reduce the diffusion of dopants. The defects caused by the ion implantation process **210** may be reduced by other subsequent processes. In an embodiment, the temperature T_2 may be between about 700° C. and about 1000° C. If the temperature T_2 is below 700° C., the pre-bake process **218** may not be able to remove the undesirable components (i.e., oxide residue and/or unintentionally

formed oxidized layer) on the top surface **202t** of the substrate **202**. That is, the temperature **T2** is greater than the desorption temperature of the pad oxide layer **204** and the desorption temperature of the unintentionally formed oxidized layer. If the temperature **T2** is above 1000° C., too many dopants in the n-type doped region **202N** and p-type doped region **202P** may diffuse into adjacent regions, reducing the dopant concentration in the mesa region and thus disadvantageously increasing the leakage current. Considering the thermal budget and the pre-bake temperature **T2** of the pre-bake process **218**, the pre-bake temperature **T2** may be maintained for a duration **t2** that may be greater than the duration **t1** of the anneal process **212**. In an embodiment, the duration **t2** of the pre-bake process **218** may be between several seconds and hundred seconds, and the anneal process **212** may include a spike anneal that may be performed for less than one second. In embodiments, the duration **t2** of the pre-bake process **218** is less than 300 seconds. In an embodiment, **t2** is between about 80 seconds and about 120 seconds to evaporate the solvent in the substrate **202** while providing the mesa region (e.g., mesa region **500T** shown in FIG. 17A) a satisfactory doping profile, thereby reducing the leakage current of the workpiece **200**.

[0030] The pre-bake process **218** may be performed in the second process chamber **216b** under a pressure that is less than 300 Torr. If the pressure is greater than 300 Torr, the efficiency of the removal of the undesirable components (e.g., oxide) and thus the quality of semiconductor layers that would be formed on the substrate **202** may be reduced. One or more gases, such as helium (He), nitrogen (N.sub.2), argon (Ar), hydrogen (H.sub.2), combinations thereof, and/or other suitable gases may be provided in the second process chamber **216b** with a flow rate of about 0.1 standard liter per minute (slm) to about 50 slm. It is noted that, the gas environment in the second process chamber **216b** is substantially free of oxygen, thereby preventing undesired oxidation of the substrate **202** during the pre-bake process **218**. That is, a partial pressure of oxygen may be substantially equal to 0. In an embodiment, the gas environment in the second process chamber **216b** is substantially filled by hydrogen (H₂) to accelerate the removal of the pad oxide layer (e.g., the first hard mask layer **204**) and/or the unintentionally formed oxidized layer on the top surface **202t** of the substrate **202**. In some embodiments, during the pre-bake process **218**, the partial pressure of H₂ in the second process chamber **216b** is greater than a partial pressure of H₂ associated with the anneal process **212**. After the pre-bake process **218**, a doping profile of the n-type doped region **202N** is changed due to thermal diffusion caused by the pre-bake process **218**. A doping profile of the n-type doped region **202N** after the pre-bake process **218** is shown in FIG. 21 and represented by curve **430**.

[0031] Referring to FIGS. 1, 7C, 8A and 8B, method **100** includes a block **112** where a vertical stack **220** of alternating sacrificial layers **222** and channel layers **224** are deposited on the substrate **202**. After performing the pre-bake process **218**, the workpiece **200** may still be positioned in the second process chamber **216b** and semiconductor layers (e.g., semiconductor layers **222** and **224** shown in FIGS. 8A-8B) may be epitaxially grown on the top surface of the substrate **202**. The vertical stack **220** is formed by depositing sacrificial layers **222** and channel layers **224** over substrate **202** in an interleaving or alternating configuration from the top surface **202t** of the substrate **202**. For example, a first one of the sacrificial layers **222** is epitaxially grown on the substrate **202**, a first one of the channel layers **224** is epitaxially grown on the first one of the sacrificial layers **222**, a second one of the sacrificial layers **222** is epitaxially grown on the first one of the channel layers **224**, and so on until the vertical stack **220** has a desired number of sacrificial layers **222** and channel layers **224**. In such embodiments, sacrificial layers **222** and channel layers **224** can be referred to as epitaxial layers. Sacrificial layers **222** and channel layers **224** may be epitaxially grown by molecular beam epitaxy (MBE), chemical vapor deposition (CVD), other suitable epitaxial growth process, or combinations thereof. It is noted that three sacrificial layers **222** and three channel layers **224** are alternately and vertically arranged as illustrated in FIGS. 8A-8B, which are for illustrative purposes only and not intended to limit the present disclosure to what is explicitly illustrated therein. It is understood that any number of sacrificial layers **222** and

channel layers **224** can be formed in the stack **220**. In some embodiments, the number of the channel layers **224** is between 2 and 10.

[0032] Each channel layer **224** may include a semiconductor material such as, silicon, germanium, silicon carbide, silicon germanium, GeSn, SiGeSn, SiGeCSn, other suitable semiconductor materials, or combinations thereof, while each sacrificial layer **222** has a composition different from that of the channel layer **224**. In an embodiment, the channel layer **224** includes silicon (Si), the sacrificial layer **222** includes silicon germanium (SiGe). In an embodiment, the sacrificial layers **222** and the channel layers **224** are formed by a selective CVD process **226** that introduces a silicon-containing precursor and a germanium-containing precursor and a carrier gas into the second process chamber **216b**, where the silicon-containing precursor and the germanium-containing precursor interact with semiconductor surfaces of the workpiece **200** to form the sacrificial layers **222** and the channel layers **224**, respectively. The silicon-containing precursor includes SiH.sub.4, Si.sub.2H.sub.6, SiH.sub.2Cl.sub.2, SiHCl.sub.3, SiCl.sub.4, other suitable silicon-containing precursors, or combinations thereof. The germanium-containing precursor includes GeH.sub.4, Ge.sub.2H.sub.6, GeCl.sub.4, GeCl.sub.2, other suitable germanium-containing precursors, or combinations thereof. The carrier gas may be an inert gas, such as H.sub.2, N.sub.2, H.sub.2, or Ar. In some embodiments, the sacrificial layers **222** and channel layers **224** may be epitaxially grown in a same process chamber **216b**.

[0033] In some embodiments, during the epitaxial growth of the sacrificial layers **222** and channel layers **224**, the gas environment in the second process chamber **206b** is substantially free of oxygen, thereby preventing undesired oxidation of the channel layers **224** and the sacrificial layers **222** during the selective CVD process **226**. That is, a partial pressure of oxygen may be substantially equal to 0.

[0034] In an embodiment, the selective CVD process **226** is performed in the second process chamber **216b** at temperature **T3** for a time duration **t3** and under a pressure that is less than 300 Torr to form satisfactory channel layers **224** and sacrificial layers **222**. The thermal budget of the selective CVD process **226** may be greater than the thermal budget of the pre-bake process **218**. To reduce dopants in the n-type doped region **202N**/p-type doped region **202P** from being thermally diffused and thus provide a satisfactory doping profile in the mesa region, the selective CVD process **226** is performed at temperature **T3** that is lower than the pre-bake temperature **T2**. In an embodiment, the process temperature **T3** of the selective CVD process **226** is between about 600° C. and about 800° C. When the process temperature **T3** is below 600° C., the growth rate of the epitaxial layers may be too slow. When the process temperature **T3** is above 800° C., the quality of the deposited epitaxial layers may deteriorate. The defects caused by the ion implantation process **210** may be further repaired by the selective CVD process **226** due to the thermal effect at an elevated temperature. Considering the thermal budget and the process temperature **T3** of the selective CVD process **226** and the number of semiconductor layers that would be formed in the stack **220**, the process temperature **T3** may be maintained for a duration **t3** that is measured by minutes or hours and greater than the duration **t2** of the pre-bake process **218**.

[0035] A doping profile of the n-type doped region **202N** after the selective CVD process **226** is shown in FIG. **21** and is represented by curve **510**. The top portion **202NT** of the n-type doped region **202N** includes an upper region **500U** (shown in FIG. **21**), a middle region **500M**, and a lower region **500L**. A dopant concentration of the upper region **500U** is less than a dopant concentration of the middle region **500M**, and the dopant concentration of the middle region **500M** is greater than a dopant concentration of the lower region **500L**. As shown in FIG. **21**, after performing the selective CVD process **226**, the dopant concentration of the upper region **500U** is greater than 8×10^{17} atom/cm.sup.3, the dopant concentration of the middle region **500M** is greater than 8×10^{18} atom/cm.sup.3. In some embodiments, as will be described below, the upper region **500U** and the middle region **500M** would be collectively patterned to form a mesa region **500T** in the first portion **202a** and a mesa region **500T'** in the second portion **202b**. By providing the mesa

region **500T** having a doping profile (e.g., curve **510**) shown in FIG. **21**, leakage current associated with the mesa region **500T** may be reduced, thereby providing a better device performance.

[0036] Referring to FIGS. **1**, **9A** and **9B**, method **100** includes a block **114** where a second hard mask layer **228** is formed on the vertical stack **220**. In embodiments represented in FIGS. **9A** and **9B**, the second hard mask layer **228** includes a pad oxide layer **228a** and a pad nitride layer **228b** formed on the pad oxide layer **228a**. The pad oxide layer **228a** may include silicon oxide and may be formed by a second process **230**. The second process **230** may include a thermal oxidation process that has a thermal budget less than the thermal budget of the first process **206** and less than the thermal budget of the pre-bake process **218**. The second process **230** may be performed under a pressure ranging from about 1500 mTorr to about 4500 mTorr and at a process temperature **T4** ranging from about 60° C. to about 390° C. in a process environment that may include one or more reaction gases such as O.sub.2 and/or SAM24 (C.sub.8H.sub.22N.sub.2Si) for a duration **t4** of several minutes with a flow rate ranging from about 1 slm to about 10 slm. Carrier gases such as H.sub.2, N.sub.2, Ar, and/or He may be used to transport the reaction gases into a furnace or a process chamber. The process temperature **T4** is designed such that the formation of the second hard mask layer **228** will not significantly affect the mesa region's doping profile (e.g., represented by the curve **510**). The pad nitride layer **228b** may include silicon nitride and may be formed by CVD, ALD, or other suitable processes. It is understood that, in some embodiments, the process temperature for forming the pad nitride layer **228b** may be lower than the pre-bake temperature **T2** of the pre-bake process **218** to reduce dopants' thermal diffusion.

[0037] Referring to FIGS. **1**, **9A**, **9B**, **10A** and **10B**, method **100** includes a block **116** where the vertical stack **220** and an upper portion **202U** of the substrate **202** are patterned to form a number of fin-shaped active regions **232**. The patterning process may include a lithography process (e.g., photolithography or e-beam lithography) which may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. The second hard mask layer **228** may be patterned using the patterned photoresist layer as an etch mask, and the vertical stack **220** and the upper portion **200U** of the substrate **202** may be then patterned using the patterned second hard mask layer **228** as an etch mask. After the patterning, each of the fin-shaped active regions **232** includes a patterned vertical stack **220** and a patterned upper portion **202U** of the substrate **202**. The patterned upper portion **202U** in the first portion **202a** of the substrate **202** is referred to as a mesa region **500T** and the patterned upper portion **202U** in the second portion **202b** of the substrate **202** is referred to as a mesa region **500T'**. The doping profile **510** of the mesa region **500T** is shown in FIG. **21**. The doping profile of the mesa region **500T'** may be in a way similar to the doping profile **510** of the mesa region **500T** and is omitted for reason of simplicity. The mesa region **500T** and mesa region **500T'** may each have a depth (from the top surface **202t** of the substrate **202**) along the $-Z$ direction. In an embodiment, the depth may be between about 10 nm and about 20 nm to facilitate the formation of a satisfactory isolation feature between two adjacent active regions.

[0038] As shown in FIGS. **10A** and **10B**, each of the fin-shaped active regions **232** extends lengthwise along the X direction and includes channel regions **232C** and source/drain regions **232S/D**. Source/drain region(s) may refer to a source region or a drain region, individually or collectively dependent upon the context. Each channel region **232C** is disposed between two source/drain regions **232S/D**. FIG. **10B** depicts a cross-sectional view of the workpiece **200** taken along line B-B' shown in FIG. **10A**. It is noted that the number of fin-shaped active regions **232** formed over the substrate **202** and the numbers of the channel regions and source/drain regions shown in FIGS. **10A** and **10B** are for illustrative purposes only and not intended to limit the present disclosure to what is explicitly illustrated therein.

[0039] Still referring to FIGS. **1**, **10A** and **10B**, method **100** includes a block **118** where isolation structures **234** are formed to isolate two adjacent fin-shaped active regions **232**. In embodiments

represented in FIGS. 10A and 10B, the formation of the isolation structures 234 includes forming an oxide liner 234a and forming a nitride fill layer 234b over the oxide liner 234a. The formation of the oxide liner 234a may be in a way similar to the formation of the pad oxide layer 228a. For example, the oxide liner 234a may include silicon oxide and may be formed by a thermal oxidation process that has a thermal budget less than the thermal budget of the pre-bake process 218 and repeated description is omitted for reason of simplicity. The formation of the nitride fill layer 234b may be performed by using a deposition process such as a flowable CVD process 236 and may have a thermal budget greater than the thermal budget of the second process 230 (shown in FIGS. 9A-9B) to fill trenches between two adjacent fin-shaped active regions 232. The flowable CVD process 236 may be performed under 1 atmosphere (atm) and at a process temperature T5 in a furnace that may include one or more reaction gases such as H.sub.2O and N.sub.2 for a duration t5. Carrier gases such as H.sub.2, Ar, and/or He may be used to transport the reaction gases into the furnace. In this embodiment, the process temperature T5 is greater than the process temperature T4 for forming the pad oxide layer 228a and is lower than the process temperature T3 for forming the vertical stack 220 such that the performing of the flowable CVD process 236 would not significantly affect the doping profile (e.g., curve 510) of the mesa region 500T while providing the nitride fill layer 234b a satisfactory trench filling capability to provide satisfactory isolation structure 234 for the workpiece 200. That is, $T4 < T5 < T3 < T2$. In an embodiment, the process temperature T5 ranges between about 400° C. to about 800° C. Considering the thermal budget of the flowable CVD process 236, and the process temperature T5 of the flowable CVD process 236 and the height of the fin-shaped active regions 232, the duration t5 of the flowable CVD process 236 may be between about 4 and 10 hours and is greater than the duration t2 of the pre-bake process 218.

[0040] In some embodiments, the isolation structure 234 may be a single-layer structure and may be formed of any suitable materials such as tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), and may be formed by any suitable method such as CVD, spin-on-glass (SOG), other suitable methods, or combinations thereof.

[0041] After forming the nitride fill layer 234b over the oxide liner 234a, a chemical-mechanical planarization/polishing (CMP) process may be performed until the top surface of the hard mask layer 228 is exposed. The oxide liner 234a and the nitride fill layer 234b are then etched back to form the isolation structure 234 and a trench 238 that exposes a top surface of the isolation structure 234 and sidewall surfaces of two adjacent fin-shaped active regions 232.

[0042] Referring to FIGS. 1, 11A and 11B, method 100 includes a block 120 where cladding layers 240 are formed over the workpiece 200 and extending along sidewall surfaces of each of the fin-shaped active regions 232. In the present embodiments, the cladding layers 240 may have a composition substantially the same as that of the sacrificial layer 222, such that they may be selectively removed by a common etching process. In the present embodiment, the cladding layers 240 are formed of SiGe. In some embodiments, the formation of the cladding layers 240 may include conformally depositing a material layer over surfaces of the workpiece 200 to partially fill the trench 238 and performing an anisotropic etching process to selectively remove portions of the material layer that are not extending along sidewalls of the fin-shaped active regions 232, thereby exposing portions of the isolation structure 234 and a top surface of the hard mask layer 228.

[0043] Still referring to FIGS. 1, 11A and 11B, method 100 includes a block 122 where a hybrid fin 241 is formed between two adjacent cladding layers 240 to substantially fill the trench 238. In an embodiment, the hybrid fin 241 includes a dummy fin 242 and a helmet layer 244 formed on the dummy fin 242. The dummy fin 242 is a dielectric feature and may be a multi-layer structure or a single layer structure and may include silicon nitride, silicon carbonitride (SiCN), silicon oxycarbonitride (SiOCN), silicon oxide, silicon carbide, FSG, combinations thereof, or other suitable dielectric materials and may be formed by any suitable deposition process. A top surface of

the dummy fin **242** may be coplanar with a top surface of the topmost channel layer **224**. The helmet layer **244** may be a high-k dielectric layer and may include aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, other high-k material, or a suitable dielectric material and may be deposited by a CVD process, an ALD process, a PVD process, and/or other suitable process. The workpiece **200** is then planarized using a CMP process to remove excess portions of the helmet layer **244** on the hard mask layer **228**. In some implementations, the helmet layer **244** may be configured to isolate two adjacent gate structures.

[0044] Referring to FIGS. **1**, **12A**, **12B**, **13A** and **13B**, method **100** includes a block **124** where a dummy gate structure **246** is formed over channel regions **232C** of the fin-shaped active regions **232**. As shown in FIGS. **12A** and **12B**, after forming the helmet layer **244**, the workpiece **200** is etched to selectively remove the hard mask layer **228** and a portion of the cladding layer **240** that extends along the sidewalls of the hard mask layer **228** without substantially etching the helmet layer **244** or the topmost channel layer **224**. In some implementations, the etching process employed in block **124** may include a selective dry etching process. In some implementations, the etching process may include a selective wet etching process (e.g., selective to SiGe) that includes ammonium hydroxide (NH₄OH), hydrogen fluoride (HF), hydrogen peroxide (H₂O₂), or a combination thereof. After the etching process, the cladding layers **240** and the topmost channel layers **224** are substantially coplanar.

[0045] In embodiments represented in FIGS. **13A** and **13B**, a dummy gate structure **246** is then formed over channel regions **232C** of the fin-shaped active regions **232**. In this embodiment, a gate replacement process (or gate-last process) is adopted where the dummy gate structure **246** serves as a placeholder for a functional gate stack. Other processes and configurations are possible. While not explicitly shown, the dummy gate structure **246** may include a dummy dielectric layer and a dummy electrode disposed over the dummy dielectric layer. In some embodiments, the dummy dielectric layer may include silicon oxide and the dummy electrode may include polycrystalline silicon (polysilicon). A gate-top mask **248** is formed on the dummy gate structure **246**. After forming the dummy gate structure **246**, a gate spacer **250** is formed along sidewalls of the dummy gate structure **246** and the gate-top mask **248**. Dielectric materials for the gate spacer **250** may be selected to allow selective removal of the dummy gate structure **246** without substantially damaging the gate spacer **250**. The gate spacer **250** may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, silicon oxynitride, and/or combinations thereof.

[0046] Referring to FIGS. **1**, **14A** and **14B**, method **100** includes a block **126** where source/drain regions **232S/D** of the fin-shaped active regions **232** are recessed to form source/drain trenches **252**. With the dummy gate structure **246** and the gate spacer **250** serving as an etch mask, the source/drain regions **232S/D** of the fin-shaped active regions **232** and a portion of the cladding layers **240** not covered by the dummy gate structure **246** are removed by an anisotropic etching process to form source/drain trenches **252**. The anisotropic etching process in block **126** may include a dry etching process and may implement hydrogen, a fluorine-containing gas (e.g., CF₄, SF₆, CH₂F₂, CHF₃, and/or C₂F₆), a chlorine-containing gas (e.g., Cl₂, CHCl₃, CCl₄, and/or BCl₃), a bromine-containing gas (e.g., HBr and/or CHBr₃), an iodine-containing gas, other suitable gases and/or plasmas, and/or combinations thereof. Source/drain trenches **252** may not only extend through the stack **220**, but also extend through a portion of the substrate **202**.

[0047] Still referring to FIGS. **1**, **14A** and **14B**, method **100** includes a block **128** where inner spacer features **253** are formed in the workpiece **200**. After forming the source/drain trenches **252**, the sacrificial layers **222** exposed in the source/drain trenches **252** are selectively and partially recessed to form inner spacer recesses (filled by inner spacer features **253**), while the exposed channel layers **224** are substantially unetched. In some embodiments, this selective recess may

include a selective isotropic etching process (e.g., a selective dry etching process or a selective wet etching process), and the extent at which the sacrificial layers **222** is recessed is controlled by duration of the etching process. After the formation of the inner spacer recesses, an inner spacer material layer is then conformally deposited using CVD or ALD over the workpiece **200**, including over and into the inner spacer recesses. The inner spacer material may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, or silico oxynitride. After the deposition of the inner spacer material layer, the inner spacer material layer is etched back to form inner spacer features **253**.

[0048] Referring to FIGS. **1**, **15A** and **15B**, method **100** includes a block **130** where epitaxial source/drain features **254N** and **254P** are formed in the source/drain trenches **252**. Source/drain feature(s) may refer to a source or a drain, individually or collectively dependent upon the context. N-type source/drain features **254N** are formed in source/drain trenches **252** over the second portion **202b** and p-type source/drain features **254P** are formed in source/drain trenches **252** over the first portion **202a**. The n-type source/drain features **254N** and the p-type source/drain features **254P** each may be epitaxially and selectively formed from exposed top surfaces of the substrate **202** and exposed sidewalls of the channel layers **224** by using an epitaxial process, such as VPE, UHV-CVD, MBE, and/or other suitable processes. The n-type source/drain features **254N** are coupled to the channel layers **224** over the second portion **202b** and may include silicon, phosphorus-doped silicon, arsenic-doped silicon, antimony-doped silicon, or other suitable material and may be in-situ doped during the epitaxial process by introducing an n-type dopant, such as phosphorus, arsenic, or antimony, or ex-situ doped using a junction implant process. The p-type source/drain features **254P** are coupled to the channel layers **224** over the first portion **202a** and may include germanium, gallium-doped silicon germanium, boron-doped silicon germanium, or other suitable material and may be in-situ doped during the epitaxial process by introducing a p-type dopant, such as boron or gallium, or ex-situ doped using a junction implant process.

[0049] Referring to FIGS. **1**, **16A**, **16B**, **17A** and **17B**, method **100** includes a block **132** where the dummy gate structure **246** is replaced by functional gate stacks (e.g., gate stack **265**). After forming the source/drain features **254N** and **254P**, a contact etch stop layer (CESL) **256** and an interlayer dielectric (ILD) layer **258** are deposited over the workpiece **200**. The CESL **256** may include silicon nitride, silicon oxynitride, and/or other suitable materials and may be formed by ALD, plasma-enhanced chemical vapor deposition (PECVD) process and/or other suitable deposition or oxidation processes. The CESL **256** may be deposited on top surfaces of the source/drain features **254N-254P** and sidewalls of the gate spacer **250**. The ILD layer **258** is deposited by a PECVD process or other suitable deposition technique over the workpiece **200** after the deposition of the CESL **256**. The ILD layer **258** may include any suitable materials such as tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG)).

[0050] An etching process may be performed to selectively remove the dummy gate structure **246** to form gate trenches such as gate trench **260**. Another etching process may be then followed to selectively remove the sacrificial layers **222** without substantially removing the channel layers **224** to form a number of gate openings **262**. In the present embodiments, the etching process in this channel release process also removes the cladding layer **240** previously covered by the dummy gate structure **246**. After forming the gate trenches (e.g., gate trench **260**) and gate openings **262**, a gate stack **265** is formed in the gate trench **260** and the gate openings **262** to wrap around each of the channel members **224** over the first portion **202a**. Similarly, a gate stack may be formed over the workpiece **200** to wrap around each of the channel members **224** over the second portion **202b**. Each of the gate stacks include an interfacial layer. In some embodiments, the interfacial layer may include silicon oxide. A gate dielectric layer is then deposited over the interfacial layer using ALD, CVD, and/or other suitable methods. The gate dielectric layer may include high-k dielectric materials. As used herein, high-k dielectric materials include dielectric materials having a high

dielectric constant, for example, greater than that of thermal silicon oxide (~3.9). In one embodiment, the gate dielectric layer may include hafnium oxide. Alternatively, the gate dielectric layer may include other high-k dielectrics, such as titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₅), hafnium silicon oxide (HfSiO₄), zirconium oxide, zirconium silicon oxide (ZrSiO₂), lanthanum oxide (La₂O₃), aluminum oxide (Al₂O₃), yttrium oxide (Y₂O₃), SrTiO₃(STO), BaTiO₃(BTO), BaZrO, hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), (Ba,Sr)TiO₃(BST), silicon nitride (SiN), silicon oxynitride (SiON), combinations thereof, or other suitable material. A gate electrode layer is then deposited over the gate dielectric layer. The gate electrode layer may be a multi-layer structure that includes at least one work function layer and a metal fill layer. By way of example, the gate stack formed over the second portion **202b** may include an n-type work function metal layer such as Ti, Al, Ag, Mn, Zr, TiAl, TiAlC, TaC, TaCN, TaSiN, TaAl, TaAlC, TiAlN, other n-type work function material, or combinations thereof, and the gate stack **265** may include a p-type function metal layer such as TiN, TaN, Ru, Mo, Al, WN, ZrSi₂, MoSi₂, TaSi₂, NiSi₂, WCN, other p-type work function material, or combinations thereof.

[0051] Referring to FIGS. **1**, **17A** and **17B**, method **100** includes a block **134** where further processes may be performed to complete the fabrication of the semiconductor device **200**. For example, method **100** may further include recessing the gate stacks, forming dielectric capping layer over the recessed gate stacks. Such further processes may also include forming an interconnect structure configured to connect the various features to form a functional circuit that includes the different semiconductor devices. The interconnect structure may include multiple interlayer dielectric (ILD) layers and multiple metal lines, contact vias, and/or power rails in each of the ILD layers. The metal lines, contact vias, and/or power rails in each ILD layer may be formed of metal, such as aluminum, tungsten, ruthenium, or copper.

[0052] In the above embodiments, parameters such as temperatures, durations, process pressures, and/or flow rates associated with various fabrication processes (e.g., the first process **206** for forming the first hard mask layer **204**, the anneal process **212**, the pre-bake process **218**, the selective CVD process **226**, the second process **230** for forming the second hard mask layer **228**, the flowable CVD process **236**) are described. The thermal budgets of those fabrication processes are determined such that the mesa region **500T** and the mesa region **500T'** each may have a satisfactory doping profile in the final semiconductor device **200**, thereby reducing leakage current. FIG. **18** illustrates a flowchart of an exemplary method **300** for determining and fine-tuning temperatures and durations of thermal treatments for performing the method of FIG. **1**, according to various embodiments of the present disclosure.

[0053] Referring to FIG. **18**, method **300** includes a block **302** where an anneal temperature **T1** and a duration **t1** for the anneal process **212** are determined. The anneal temperature **T1** is configured such that the anneal process **212** would activate dopants in the n-type doped region **202N** and/or dopants in the p-type doped region **202P** and also facilitate movement of the dopants from interstitial sites to substitutional lattice sites to reduce dislocations in crystal lattice and repair defects caused by the ion implantation process **210**. In the present embodiment, the anneal temperature **T1** is between about 950° C. and about 1200° C. The duration **t1** of the anneal process **212** may be then determined based on the thermal budget of the anneal process **212** and a whole thermal budget of the entire fabrication processes illustrated in method **100**, according to embodiments of the present disclosure.

[0054] Still referring to FIG. **18**, method **300** includes a block **304** where a process temperature **T3** and duration **t3** of the epitaxial growth (e.g., the selective CVD process **226**) for forming the stack **220** are determined. The process temperature **T3** is selected such that the deposited epitaxial layers (e.g., channel layers **224** and sacrificial layers **222**) may have satisfactory quality while being

formed in an acceptable duration to increase production. The duration t_3 of the epitaxial growth for forming the stack **220** may be then determined based on, for example, the thickness of the stack **220**, the process temperature T_3 , and the thermal budget of the epitaxial growth for forming the stack **220**. In an embodiment, the process temperature T_3 is between about 600° C. and about 800° C. To fine tune the pre-bake temperature T_2 of the pre-bake process **218** and thus provide a satisfactory doping profile for the mesa region **500T**, an initial process temperature T_3 may be set to T_{30} (e.g., 600° C.), and the initial duration t_3 may be determined accordingly. It is understood that operations in block **304** and operations in block **302** may be performed in any sequential order. For example, operations in block **304** may be performed before performing the operations in block **302**.

[0055] Referring to FIG. **18**, method **300** includes a block **306** where a pre-bake temperature T_2 and duration t_2 of the pre-bake process **218** are determined. The pre-bake temperature T_2 is initially determined such that any undesirable components (i.e., oxide residue and/or unintentionally formed oxidized layer) may be removed, the solvent in the substrate **202** may be evaporated, and the defects caused by the anneal process **212** may be significantly reduced. That is, the pre-bake temperature T_2 is no less than the desorption temperature (e.g., about 700° C.) of the unintentionally formed oxidized layer. In an embodiment, the pre-bake temperature T_2 may be initially set to T_{20} . The duration t_2 of the pre-bake process **218** may be determined based on a thermal budget of the pre-bake process **218** and the initial pre-bake temperature T_2 .

[0056] Referring to FIG. **18**, method **300** includes a block **308** where operations in blocks of method **100** are performed under the predetermined conditions (e.g., the initial temperatures T_{20} , T_{30} and corresponding durations of the pre-bake process **218** and the selective CVD process **226**) to fabricate a first test device. The first test device may be in a way similar to the semiconductor device **200** except for the doping profile of the mesa region. More specifically, increasing the pre-bake temperature T_2 would repair more defects caused by the anneal process, thereby reducing leakage current. However, increasing the pre-bake temperature T_2 would also increase dopants' thermal diffusion and lead to a decreased dopant concentration in the mesa region, resulting an increased leakage current.

[0057] Referring to FIG. **18**, method **300** includes a block **310** where a doping profile of the mesa region of the first test device ("test doping profile") is obtained and the leakage current associated with the mesa region of the first test device ("test leakage current") is evaluated to determine whether the test doping profile is a satisfactory doping profile. In situations where the leakage current is greater than a predetermined threshold, the test doping profile is not a satisfactory doping profile. If the test doping profile is not a satisfactory doping profile, method **300** moves to block **314** where the pre-bake temperature T_2 of the pre-bake process **218** is reduced from the initial pre-bake temperature T_{20} to an adjusted pre-bake temperature T_2' . After reducing the initial pre-bake temperature, method **300** moves to block **316** where the duration of the pre-bake process **218** is prolonged from the initial duration to an adjusted duration, based on the thermal budget of the pre-bake process **218** and the adjusted pre-bake temperature. After determining the adjusted pre-bake temperature T_2' and adjusted duration for the pre-bake process **218**, method **300** moves to block **318** where the process temperature T_3 of the epitaxial growth (e.g., the selective CVD process **226**) for forming the stack **220** is adjusted. Since the pre-bake temperature T_2 is reduced, the number of defects that would be repaired by the pre-bake process **218** may be reduced. To reduce the total number of defects in the final semiconductor device **200**, the process temperature T_3 of the epitaxial growth for forming the stack **220** may be increased from T_{30} to T_3' to repair more defects. It is understood that, after adjusting, the process temperature T_3' would still be between about 600° C. and about 800° C. For example, an initial process temperature T_{30} may be about 600° C., and after the adjusting, the process temperature T_3' may be about 640° C. After determining the adjusted process temperature T_3' for the epitaxial growth for forming the stack **220**, method **300** moves to block **320** where the growth duration of the epitaxial growth may be

determined (e.g., shortened) accordingly without substantially changing the thermal budget of the epitaxial growth process.

[0058] After adjusting the temperatures and durations of the pre-bake process **218** and the selective CVD process **226**, method **300** loops back to block **308** where operations in blocks of method **100** are performed under the adjusted fabrication parameters (e.g., the adjusted temperatures T_2' , T_3' , and durations) to fabricate a second test device. The doping profile of the mesa region of the second test device is then evaluated. If the doping profile of the mesa region of the second test device is a satisfactory doping profile, method **300** moves to block **312** where the adjusted fabrication parameters (e.g., the adjusted temperatures and durations) used to fabricate the second test device may be used in actual fabrication of semiconductor devices (e.g., the semiconductor device **200**). Otherwise, the method **300** may loop back to the blocks **314-320** and the temperatures and durations of the pre-bake process **218** and the selective CVD process **226** may be further adjusted.

[0059] FIG. **19A** illustrates a bar chart showing the relationships among parameters of thermal treatments for performing the method of FIG. **1** and FIG. **19B** illustrates a table showing ranges of the parameters of thermal treatments for performing the method of FIG. **1**, according to various embodiments of the present disclosure. More specifically, relationships of the process temperatures, process durations, process pressures and gas flow rates of the first process **206**, anneal process **212**, pre-bake process **218**, selective CVD process **226**, the second process **230** and the flowable CVD process **236** are illustrated in FIG. **19A**. Some of the relationships, for example, the relationships among those process temperatures and relationships among those process durations, may be obtained by method **300**. It is understood that the satisfactory doping profile (e.g., doping profile (e.g., curve **510**) shown in FIG. **21**) of the mesa region **500T** of the semiconductor device **200** is a function of the thermal budgets as well as the respective temperatures and durations of the anneal process **212**, the pre-bake process **218**, the epitaxial growth process (e.g., the selective CVD process **226**) and a function of the thermal budget of a rest of fabrication processes (e.g., flowable CVD process **236**) performed after the performing of the epitaxial growth process. Detailed various relationships among those parameters (e.g., temperature, duration, process pressure, flow rate) in various thermal treatments are shown in FIG. **19A** and detailed ranges of those parameters are shown in FIG. **19B**, and repeated description is omitted for reason of simplicity.

[0060] FIG. **20** illustrates doping profiles of the n-type doped region **202N** of the workpiece **200** before and after the performing of the anneal process **212**. Curve **410** represents the doping profile of the n-type doped region **202N** before the performing of the anneal process **212**, and curve **420** represents the doping profile of the n-type doped region **202N** after the performing of the anneal process **212**. In embodiments represented in FIG. **20**, before the anneal process **212**, a dopant concentration of a top portion of the n-type doped region **202N** (i.e., the top portion **202NT**) is higher than a dopant concentration of a bottom portion of the n-type doped region **202N** (i.e., the bottom portion **202NP**). After the anneal process **212**, the dopant concentration of the top portion **202NT** is reduced comparing to that of the top portion **202NT** before the anneal process **212**, the dopant concentration of the bottom portion **202NP** is increased comparing to that of the bottom portion **202NP** before the anneal process **212**, and, after the anneal process **212**, the dopant concentration of the top portion **202NT** is still higher than the dopant concentration of the bottom portion **202NP**. In embodiments represented in FIG. **20**, the top portion **202NT** refers to the region of the n-type doped region **202N** that ranges from the top surface **202t** of the substrate **202** to a depth of about 35 nm in the n-type doped region **202N**, and the bottom portion **202NB** is a rest of the n-type doped region **202N**.

[0061] FIG. **21** illustrates a curve **430** representing a doping profile of the top portion **202NT** after the performing of the pre-bake process **218**. A portion **510** of the curve **430** (“the curve **510**”) represents a doping profile of an upper portion (i.e., the mesa region **500T**) of the top portion **202NT** after the performing of the epitaxial grow process (e.g., the selective CVD process **226**). In

embodiments represented in FIG. 21, after the selective CVD process 226, the dopant concentration of the mesa region 500T is higher than 8×10^{17} atom/cm.^{sup.3} and is not substantially uniform throughout its depth along the Z axis. More specifically, the mesa region 500T includes a region 500U near the top surface 202t of the substrate and a region 500M under the region 500U. A dopant concentration of the region 500U is higher than 8×10^{17} atom/cm.^{sup.3}, and a dopant concentration of the region 500M is greater than a dopant concentration of the region 500U and is higher than 8×10^{18} atom/cm.^{sup.3}. By providing the mesa region such a doping profile, leakage current associated with the mesa region 500T may be advantageously reduced, thereby providing better device performance.

[0062] Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, the present disclosure provides a method of forming a semiconductor device that has a satisfactory doping profile in the mesa region of the semiconductor device. Thermal budgets, process temperatures and process durations are configured such that unrepaired defects caused by ion-implantation process and dopant diffusions caused by various thermal treatments are both in acceptable ranges, thereby reducing leakage current without yielding significant penalty. Embodiments of the disclosed methods can be readily integrated into existing processes and technologies for manufacturing GAA FETs.

[0063] The present disclosure provides for many different embodiments. Semiconductor devices and methods of fabrication thereof are disclosed herein. In one exemplary aspect, the present disclosure is directed to a method. The method includes providing a workpiece comprising a first hard mask layer on a top surface of a substrate, performing an ion implantation process to form a doped region in the substrate, after the performing of the ion implantation process, annealing the workpiece at temperature T1, selectively removing the first hard mask layer, after the selectively removing of the first hard mask layer, performing a pre-bake process at temperature T2, and after the performing of the pre-bake process, epitaxially growing a vertical stack of alternating channel layers and sacrificial layers on the substrate. The temperature T2 is lower than the temperature T1.

[0064] In some embodiments, the epitaxially growing of the vertical stack may be performed at temperature T3, wherein T3 is lower than T2. In some embodiments, the performing of the pre-bake process may include placing the workpiece in a first chamber of a chemical vapor deposition (CVD) tool, and the epitaxially growing of the vertical stack may be performed in the first chamber. In some embodiments, the performing of the pre-bake process may include exposing the substrate to a first gas environment in the first chamber to remove a residue of the first hard mask layer, the first hard mask layer may include an oxide layer, and the first gas environment may include hydrogen (H.sub.2). In some embodiments, the annealing of the workpiece may include exposing the substrate to a second gas environment that includes hydrogen (H.sub.2), the hydrogen in the second gas environment may have a partial pressure that is different from a partial pressure of the hydrogen in the first gas environment. In some embodiments, the method may also include, after the epitaxially growing of the vertical stack, forming a second hard mask layer over the vertical stack at temperature T4, and the temperature T4 may be lower than the temperature T1 and the temperature T2. In some embodiments, the method may also include patterning the vertical stack and an upper portion of the substrate to form fin-shaped active regions, each of the fin-shaped active regions may include a patterned vertical stack formed over a mesa region of the substrate, forming an isolation structure to isolate two adjacent fin-shaped active regions, forming a dummy gate structure over channel regions of the fin-shaped active regions, forming source/drain features in and over source/drain regions of the fin-shaped active regions, each of the channel regions being disposed between two source/drain features, selectively removing the dummy gate structure and the sacrificial layers, and forming a gate stack to wrap around each of the channel layers. In some embodiments, after the forming of the gate stack, a dopant concentration of the mesa region may be greater than 8×10^{17} atom/cm.^{sup.3}. In some embodiments, the mesa region may include an upper

portion and a lower portion, and a dopant concentration of the lower portion may be greater than a dopant concentration of the upper portion.

[0065] In another exemplary aspect, the present disclosure is directed to a method. The method includes forming a dielectric layer on a substrate, after the forming of the dielectric layer, forming a doped region in the substrate, performing an anneal process to the substrate at temperature T1 for a duration t1, after the performing of the anneal process, selectively removing the dielectric layer, after the selectively removing of the dielectric layer, pre-baking the substrate at temperature T2 for a duration t2, and forming a vertical stack of alternating epitaxial semiconductor layers on the substrate at temperature T3 for a duration t3, where the temperature T2 is lower than the temperature T1 and is greater than the temperature T3, and the duration t2 is less than the duration t3 and is greater than the duration t1.

[0066] In some embodiments, the temperature T1 may be between about 950° C. and about 1200° C. In some embodiments, the temperature T2 may be between about 700° C. and about 1000° C. In some embodiments, the temperature T3 may be between about 600° C. and about 800° C. In some embodiments, the anneal process may be performed in a first gas environment, the pre-baking of the substrate may be performed in a second gas environment, and a partial pressure of oxygen in the second gas environment may be less than a partial pressure of oxygen in the first gas environment. In some embodiments, the forming of the doped region may include performing an ion implantation process.

[0067] In some embodiments, the method may include patterning the vertical stack and a top portion of the substrate to form an active region, the active region comprising a patterned vertical stack formed on a patterned top portion of the substrate, forming a gate structure over a channel region of the active region, and forming source/drain features over source/drain regions of the active region. After the patterning, a dopant concentration of the patterned top portion of the substrate may be greater than 8E17 atom/cm³. In some embodiments, before the performing of the anneal process, the doped region includes an upper portion having a first dopant concentration and a lower portion having a second dopant concentration different from the first dopant concentration, and after the performing of the anneal process, the first dopant concentration may decrease, and the second dopant concentration may increase.

[0068] In yet another exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes forming an oxide dielectric layer on a substrate, performing an ion implantation process to form a doped region in the substrate, performing a first thermal treatment to the substrate in a first gas environment and at temperature T1 for a duration t1, after the performing of the first thermal treatment, selectively removing the oxide dielectric layer, after the selectively removing of the oxide dielectric layer, performing a second thermal treatment to the substrate in a second gas environment and at temperature T2 for a duration t2, and epitaxially forming a vertical stack of semiconductor layers on the substrate in a third gas environment and at temperature T3 for a duration t3. The temperature T2 is lower than the temperature T1 and is greater than the temperature T3, the duration t2 is less than the duration t3 and is greater than the duration t1, and a partial pressure of hydrogen in the second gas environment is different than a partial pressure of hydrogen in the first gas environment.

[0069] In some embodiments, a partial pressure of oxygen in the second gas environment may be substantially equal to 0. In some embodiments, the temperature T2 may be greater than a desorption temperature of the oxide dielectric layer.

[0070] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a channel region over a substrate, the channel region comprising a plurality of nanostructures over a base region protruded from the substrate; an isolation feature disposed over the substrate and adjacent to a sidewall of the base region; a gate structure over the channel region, and a gate isolation structure adjacent to the gate structure and comprising a first layer and a second layer over the first layer, wherein a dielectric constant of the first layer is different from a dielectric constant of the second layer wherein the base region comprises an upper portion and a lower portion, a dopant concentration of the upper portion is greater than 8×10^{17} atom/cm.³, and a dopant concentration of the lower portion is greater than the dopant concentration of the upper portion.
2. The semiconductor device of claim 1, wherein, the dopant concentration of the lower portion is greater than 8×10^{18} atom/cm.³.
3. The semiconductor device of claim 1, wherein a depth of the base region is between about 10 nm and about 20 nm.
4. The semiconductor device of claim 1, wherein the substrate comprises a doped region disposed immediately under the lower portion of the base region, and the dopant concentration of the lower portion is greater than a dopant concentration of the doped region.
5. The semiconductor device of claim 1, further comprising: source/drain features electrically coupled to the plurality of nanostructures, wherein the source/drain features overhang the isolation feature.
6. The semiconductor device of claim 5, further comprising: inner spacers disposed between the source/drain features and a lower portion of the gate structure; and gate spacers extending along sidewall surfaces of an upper portion of the gate structure.
7. The semiconductor device of claim 6, wherein a top surface of the gate isolation structure is above a top surface of a topmost nanostructure of the plurality of nanostructures.
8. The semiconductor device of claim 7, wherein a width of the gate isolation structure is less than a width of the isolation feature.
9. The semiconductor device of claim 1, wherein a top surface of the isolation feature is below a top surface of the base region.
10. A semiconductor device, comprising: a first base fin and a second base fin over a substrate; a first plurality of nanostructures over the first base fin, a second plurality of nanostructures over the second base fin, an isolation feature over the substrate and interfacing sidewalls of the first base fin and the second base fin; a first source/drain feature coupled to the first plurality of nanostructures, a portion of the first source/drain feature overhanging the isolation feature; a second source/drain feature coupled to the second plurality of nanostructures, a portion of the second source/drain feature overhanging the isolation feature; and a gate structure wrapping around the first plurality of nanostructures and the second plurality of nanostructures, wherein each of the first base fin and the second base fin comprises an upper portion over a lower portion, wherein a dopant concentration of the lower portion is greater than the dopant concentration of the upper portion.
11. The semiconductor device of claim 10, wherein the first source/drain feature and the second source/drain feature comprise different types of dopants.
12. The semiconductor device of claim 10, wherein a top surface of the isolation feature is below a top surface of the first base fin and a top surface of the second base fin.
13. The semiconductor device of claim 10, wherein the dopant concentration of the upper portion is greater than 8×10^{17} atom/cm.³.
14. The semiconductor device of claim 10, wherein the dopant concentration of the lower portion is

greater than 8×10^{18} atom/cm.³.

15. The semiconductor device of claim 10, wherein a depth of the first base fin is between about 10 nm and about 20 nm.

16. The semiconductor device of claim 15, further comprising: inner spacers providing isolation between the gate structure and the first source/drain feature.

17. A semiconductor device, comprising: an active region comprising a channel region and a source/drain region, wherein the channel region comprises a base fin extending from a substrate and a plurality of nanostructures over the base fin, wherein the base fin comprises an upper portion over a lower portion, wherein a dopant concentration of the lower portion is greater than the dopant concentration of the upper portion; an isolation feature disposed alongside the active region; a gate structure disposed over the channel region; a gate spacer disposed along a sidewall of the gate structure; a source/drain feature disposed over the source/drain region; a contact etch stop layer (CESL) disposed over the source/drain feature; and an interlayer dielectric (ILD) layer disposed over the CESL, wherein a portion of the CESL extends along a sidewall of the gate spacer such that the gate spacer is between the gate structure and the CESL.

18. The semiconductor device of claim 17, wherein the dopant concentration of the upper portion is greater than 8×10^{17} atom/cm.³.

19. The semiconductor device of claim 18, wherein the dopant concentration of the lower portion is greater than 8×10^{18} atom/cm.³.

20. The semiconductor device of claim 18, wherein a depth of the base fin is between about 10 nm and about 20 nm.
